

Semester: **Monsoon**

Examination: **END SEM EXAM, B. Tech (Mining Engineering)**

Subject: **SURFACE MINING (MNC 300)**

Instruction: Answer **ALL** questions

Session: **2024-25**

Semester: **V**

Max. Marks: **100**

Time: **3 Hours**

Assume relevant data wherever necessary and mention them

Q.No.	Question	Marks
1.	Prepare a statement of heavy earth moving machinery required for a surface iron ore mine to be worked by shovel-dumper combination to produce 5.0 million tones of ore per annum with an average stripping ratio of 0.1 tonne/tonne. The mine has been planned to be worked with 12 m high benches. Average haul distance (one way) for both ore and waste is 1 km.	15
2.	Estimate the number of drills required to achieve a production of 10 million ton/year of coal from a surface mine having coal seam of thickness 10 m. The coal seam is flat and horizontal and the stripping ratio of the mine is $3m^3/ton$	15
3.	a) Derive the effective productivity of a loading shovel. b) In case of a shovel-dumper operation in a surface mine, prove that the effective productivity of shovel depends on the dumper capacity in case of single spotting condition only, but not in double spotting condition.	15
4.	a) How is the bucket capacity of a dragline estimated? b) Draw a balancing diagram for simple side casting operation of a dragline and derive the expression, in terms of known parameter, for i. height of spoil heap ii. reach of the dragline c) What is meant by MUF of a dragline and what is its physical significance?	15
5.	Explain with the help of suitable sketches the (i) full bench method, (ii) block mining method and (iii) step-cut method of continuous surface miner operation. Also mention the applicability and limitations of each of the methods. OR a) With the help of suitable sketches, describe lateral block cut method operation of bucket wheel excavator (BWE). b) Derive the theoretical hourly output for a bucket wheel excavator. c) Calculate the hourly productivity of the BWE from the following given data: i. Number of buckets on the wheel – 12 ii. Individual bucket capacity – 1400 lit iii. Wheel diameter – 10 m iv. Cutting speed at the knife edge – 2.2 m/sec v. % swelling of material to be excavated – 20	15
6.	a) Enlist and discuss the geometrical parameters of production benches in surface mines. How are these decided upon? b) Enlist and discuss the types of failure that may occur in highwall slopes in surface mines.	15
7	Write short notes and (or) draw labeled diagram wherever necessary for the following a) Schematic presentation of mobile in-pit crushing and conveying system b) Concept of tipping load in pay-loaders	10

Prepare a Statement of HEMM required for a Surface iron ore mine to be worked by Shovel-dumper Combination to produce 5.0 million ton of ore per annum with an average stripping ratio of 0.1 tonne/tonne. The mine has been planned to be worked by 12m high benches. Average haul distance (one way) for both ore and waste is 1 km.

Given Conditions

1. ore production per year = 5×10^6 tonnes
2. Average Stripping ratio (ton/ton) = 0.1
3. Bench height = 12m.

Assumptions and calculated Parameters

	<u>Ore</u>	<u>Waste</u>
1. Average in situ density (t/m^3)	3.6	2.8
2. Working hours per year (300×24)	7200	

Shovel Related Assumptions

1. Bucket capacity (B_c), m^3	5
2. Bucket fill factor (B_f),	0.8
3. % Swell (S)	25%
4. Swell factor (S_f) $\frac{1m}{1m+S}$	0.80
5. Time Cycle (t_c) sec.	30
6. Factor of swing (F_s)	1
7. Overall utilisation factor (U_s)	0.56

Dumper Related Assumptions

1. Rated capacity of dumper (te)	50
2. Spotting time (min)	0.3
3. Dumping time (min)	1.5
4. Load travel speed (km/hr)	20
5. Empty travel speed (km/hr)	30
6. Haulage distance (km)	1

Calculation

$$\text{Iron ore production} = 5,000,000$$

$$\begin{aligned}\text{Handling of waste/yr (te)} &= SR \times P \\ &= 0.1 \times 5 \times 10^6 \\ &= 500,000\end{aligned}$$

$$\text{Total material handling} = \text{55,00,000} \text{ per year}$$

$$\begin{aligned}\text{Average density of composite} &= \frac{(1+SR) \rho_o \rho_w}{\rho_w + \rho_o SR} \\ &= 3.5088 \\ &\approx 3.509\end{aligned}$$

Calculating No. of shores

$$\begin{aligned}\text{Required material handling (yr)} &= 55,00,000 \\ \text{Avg density Composite} &= 3.509\end{aligned}$$

$$\begin{aligned}\text{Effective production per} \\ \text{Shore/yr (te)} &= \frac{B_c \times B_f \times S_f \times 3600}{t_c \times F_s} E \times H \times P \\ &= 54,32,942 \text{ (Integer)}\end{aligned}$$

$$\begin{aligned}\therefore \text{No. Shores required} &= \frac{55,00,000}{54,32,942} \\ &= 1.012\end{aligned}$$

$$\text{Actual No. of shores required} \approx 2.$$

Calculation for number of Dumpers

Effective dumper capacity = 50 te.

tonne per pass of shovel bucket

$$= B_c \times B_f \times S_f \times f$$

$$= 5 \times 0.8 \times 0.8 \times 3.509$$

$$= 11.229$$

No of passes reqd to fill dumper

$$= \text{Int} (50/11.229)$$

$$= 4.$$

Total loading time dumper = 4×30 (s)

$$= 2 \text{ min}$$

$$\text{spotting time} = 0.3 \text{ min.}$$

$$\text{Dumping time} = 1.5 \text{ min.}$$

$$\text{Load travel time} = \frac{60 \times 1}{20} = 3 \text{ min.}$$

$$\text{Empty travel time} = \frac{60 \times 1}{30} = 2 \text{ min.}$$

$$\text{Total cycle time dumper} = 2 + 0.3 + 1.5 + 3 + 2$$

$$= 8.8 \text{ min.}$$

$$\begin{aligned} \therefore \text{No of dumpers reqd} &= \frac{\text{total time}}{(\text{spotting time} + \text{loading time})} \\ &= \frac{8.8}{2.3} = 3.826 \end{aligned}$$

$$\text{Actual No of dumpers required} = 4$$

$$\therefore \boxed{\begin{aligned} \text{Total No of dumpers required} &= 4 \times 2 \\ &= 8. \end{aligned}}$$

Statement of HEMM

Sl. no	Item	Capacity	Quantity		Spare	Remarks
			Working	Total		
1	Shovel.	5 m ³	2	3	1	1.5 times
2.	Dumper	50 te.	8	12	4	1.5 times.
3.	Drill	150 mm.	2	3	1	1.5 times
4.	Dog	250/350 HP.	7	12	5	0.5 x shovel + 1.5 Sundry + 2 ore stock + 2 waste dump.
5.	FEL	3.2 m ³	5	8	3	1 or 2 for supporting
6.	Crane	—	1	1	2	1 to 3 for stock
7	Motor Grader	—	1	1	2	
8	Water Sprinkler	5 KL	2	3	1	

Estimate the number of drills required to achieve a production of 10 million ton/yr of Coal from Surface mine having a Coal seam of thickness 10 m. The coal seam is flat & horizontal and the stripping ratio of the mine is $3 \text{ m}^3/\text{ton}$.

Given Conditions

1. Coal production/year (te) = 100,00,000
2. Stripping ratio (m^3/te) = 3
3. Coal seam thickness (m) = 10

Assumptions & Calculated Parameters

	<u>Coal Bench</u>	<u>OB Bench</u>
1. Average insitu density	1.6	
2. Drill Dia (mm)	150	250
3. Burden (m)	4.5	6
4. Spacing (m)	6.5	8
5. Bench Height (m)	10 Given.	12 calculated
$\begin{aligned} \text{OB thickness} &= \text{Seam thickness} \times \text{SR} \times \rho \\ &= 10 \times 3 \times 1.6 \\ &= 48 \text{ m} \\ \therefore 4 \text{ Benches of } 12 \text{ m (ht)} \end{aligned}$		
	] Calculation.	
6. Length of drill rod (m)	7	10
7. Time to set drill m/c (min)	6	7
8. Time to set drill rod (min)	3	3
9. Rate of drilling (min/m)	2	4
10. Subgrade drilling.	0	100%
11. Overall utilisation.	0.6	0.6
12. Drilling hours (year Ctr)	3000	3000.

Estimating Coal Drills

a) Drilling requirement per year:

$$\left. \begin{array}{l} \text{Volume of Coal handled per} \\ \text{year} \cdot (\text{m}^3/\text{yr}) \end{array} \right\} = \frac{100,00,000}{1.6} \\ = 62,50,000$$

$$\text{Volume of Coal handled/hole} = 4.5 \times 6.5 \times 10 \\ = 292.5$$

$$\text{No of Holes reqd/year} = \frac{62,50,000}{292.5} \\ = 21367.52$$

$$\left. \begin{array}{l} \text{Actual No of Holes} \\ \text{required per year} \end{array} \right\} = 21368.$$

(b) Capacity of drill m/c

$$\text{No of feeds required} = \frac{10}{7} = 1.42$$

$$\text{Actual No of feeds} = \text{Roundup}(1.42) = 2$$

$$\therefore \text{No of feed change reqd} = 1$$

Time to drill hole

$$= \frac{\text{Setting time} + \text{rod setting time} + \text{drilling time}}{(\text{m/c})}$$

$$= 6 + 1 \times 3 + 10 \times 2 = 29 \text{ min.}$$

$$\text{Actual Holes per Hour} = \frac{E \times 60}{t}$$

$$= \frac{0.6 \times 60}{29} = 1.2414$$

$$\begin{aligned} \text{Actual Holes/yr} &= \frac{0.6 \times 60}{29} \times 3000 \\ \text{Drilled} &= 3724.14 \end{aligned}$$

$$\therefore \text{No of drills reqd} = \frac{\text{Requirement}}{\text{Capacity}}$$

$$= \frac{21368}{3724.14} = 5.73$$

$$\boxed{\text{Actual No of drills reqd for coal} = 6}$$

Estimation of OB drills

$$\text{Vol of OB handled/year} = \text{Prodn} \times \text{SR}$$

$$= 3,00,00,000 \text{ (m}^3\text{/yr)}$$

$$\text{Vol of OB handled per hole} = 12 \times 6 \times 8 = 576 \text{ m}^3.$$

$$\therefore \text{No of holes reqd/yr} = \frac{300000000}{576}$$

$$= 52083.33$$

$$\therefore \text{Actual No of holes reqd} = 52084$$

$$\text{Depth of drilling in OB (1 + submer) \times H}$$

$$= 13.2 \text{ m.}$$

$$\text{No of feeds reqd} = \frac{13.2}{10} = 1.32$$

$$\text{Actual No of feeds reqd} = 2.$$

$$\therefore \text{No of feed change reqd} = 1$$

$$\text{Time to drill hole} = \text{drill set time} + \text{Rod set time} + \text{drill time}$$

$$= 7 + 1 \times 3 + 4 \times 13.2$$

$$= 62.8 \text{ min.}$$

$$\therefore \text{Actual holes drilled per hr} = \frac{0.6 \times 60}{62.8}$$

$$= \frac{90}{157}$$

$$\text{Actual holes drilled per year} = \frac{90 \times 3000}{157}$$

$$= 1719.75$$

$$\therefore \text{No of OB drills reqd} = \frac{\text{Requirement}}{\text{Capacity}}$$

$$= \frac{52084}{1719.75}$$

$$= 30.286$$

$$\therefore \boxed{\text{Actual no of drills requiring for OB.} = 31 \text{ drills}}$$

(3) (a) Determine the effective productivity of a loading Shovel.

(b) In case of a shovel-dumper operation in a surface mine, prove that the effective productivity of Shovel depends on the dumper capacity in case of single spotting condition only, but not in case of double spotting.

Ans

Let

B_c = Bucket capacity of Shovel (m^3).

B_f = Bucket fill factor.

$S\%$ = % swell

$S_f = \frac{100}{100 + S} = \text{Swell factor.}$

ρ = insitu / bank density of coal or waste rock as the case may be.

t_c = Standard cycle time of shovel (s)

F_s = factor of swing.

E_s = Overall utilization factor.

H = working hours per year.

Then theoretical productivity of Shovel is given by

$$\begin{aligned} Q_{th} &= \frac{B_c \times B_f \times S_f \times 3600}{t_c \times F_s} \quad m^3(\text{bank})/\text{hr.} \\ &= \frac{B_c \times B_f \times S_f \times 3600}{t_c \times F_s} \times H \quad m^3(\text{bank})/\text{year.} \end{aligned}$$

$$= \frac{B_c \times B_f \times S_f \times 3600}{t_c \times F_s} \times \rho \quad \text{tonnes/hr.}$$

$$= \frac{B_c \times B_f \times S_f \times 3600}{t_c \times F_s} \times \rho \times H \quad \text{tonnes/year.}$$

However, the actual productivity or effective productivity (Q_{eff}) is less than the theoretical productivity

$$Q_{eff} = \frac{B_c \times B_f \times S_f \times 3600}{t_c \times F_s} \times E_s \quad (\text{m}^3/\text{hr})$$

$$= \frac{B_c \times B_f \times S_f \times 3600}{t_c \times F_s} \times E_s \times H \quad (\text{m}^3(\text{bank})/\text{year})$$

$$= \frac{B_c \times B_f \times S_f \times 3600 \times E_s \times \rho}{t_c \times F_s} \quad \text{tonnes/hr.}$$

$$= \frac{B_c \times B_f \times S_f \times 3600 \times E_s \times \rho \times H}{t_c \times F_s} \quad \text{tonnes/year.}$$

(b) Dependence of Shovel Productivity on
Dumper Capacity

Suppose, a shovel of bucket Capacity B_c is loading two dumpers of significantly different Capacity

Then effective Bucket load } = $B_L = B_c \times B_f \times S_f \times P$ te.
per pass of Shovel }

Let No of passes required } = n
to fill smaller dumper }

• No of passes required } = $m+n$ (say).
to fill bigger dumper }

For Single spotting Condition

The spotting time for both the dumpers = t_s .

Let t_c = shovel loading time (min)

∴ Effective total time to } = $(n \times t_c + t_s)$
fill the dumper }

$$\text{No of dumpers filled/hr} = \frac{60}{n \times t_c + t_s}$$

∴ Effective Hourly productivity } $P_{sd} = \frac{60 \times n \times B_L}{n \times t_c + t_s}$
of smaller dumper }

$$P_{sd} = \frac{60 \times B_L}{t_c + \left(\frac{t_s}{n}\right)} \quad \text{--- (1)}$$

Similarly for the Larger Dumper.

Effective hourly productivity of Larger Dumper } = P_{LD}

$$\begin{aligned} \text{or. } P_{LD} &= \frac{60 (m+n) B_L}{(m+n)t_c + t_s} \\ &= \frac{60 \times B_L}{t_c + \frac{t_s}{(m+n)}} \end{aligned}$$

$$\therefore \boxed{P_{LD} = \frac{60 \times B_L}{t_c + \frac{t_s}{m+n}}} \quad \text{--- (2)}$$

Comparing Eqns (1) & (2) we observe that the Numerators are same for both.

But the 2nd term of denominator in Eqn (2) smaller than that of Eqn (1) as $m+n > n$.

$$\therefore \boxed{P_{LD} > P_{SD}} \quad \text{--- (3)}$$

However, for double spotting condition

$t_s = \text{spotting time} = 0$ [for both dumpers].

$\therefore$ From Eqn (1) & (2) we get

$$\boxed{P_{LD} = P_{SD}} \quad \text{--- (4)}$$

$\therefore$ For Single spotting Condition the effective productivity of a Shovel increases with increasing dumper Capacity. However, it is independent of dumper Capacity in case of double Spotting Condition.

- (a) How is the bucket Capacity of a dragline estimated?
- (b) Draw a balancing diagram for simple sidecasting operation of a dragline and derive the expression, in terms of known parameter, for
- Height of Spoil heap .
 - Reach of dragline
- (c) What is meant by MUF of a dragline and what is its physical significance .

Estimation of Bucket Capacity of a Dragline.

For estimating the B_c , first the hourly handling is estimated from the production output (P) & the stripping ratio (SR)

$$\therefore \text{Handling requirement} = \frac{P \times SR}{365 \times 24} \text{ m}^3(\text{bank})/\text{hr}$$

$$\begin{aligned} P &= \text{Coal production per yr (te)} \\ SR &= \text{Stripping ratio (m}^3/\text{te)} \end{aligned}$$

$$\therefore \text{Handling requirement} = \frac{P \times SR}{365 \times 24} \quad \text{--- (A)}$$

The effective hourly capacity of dragline is given by

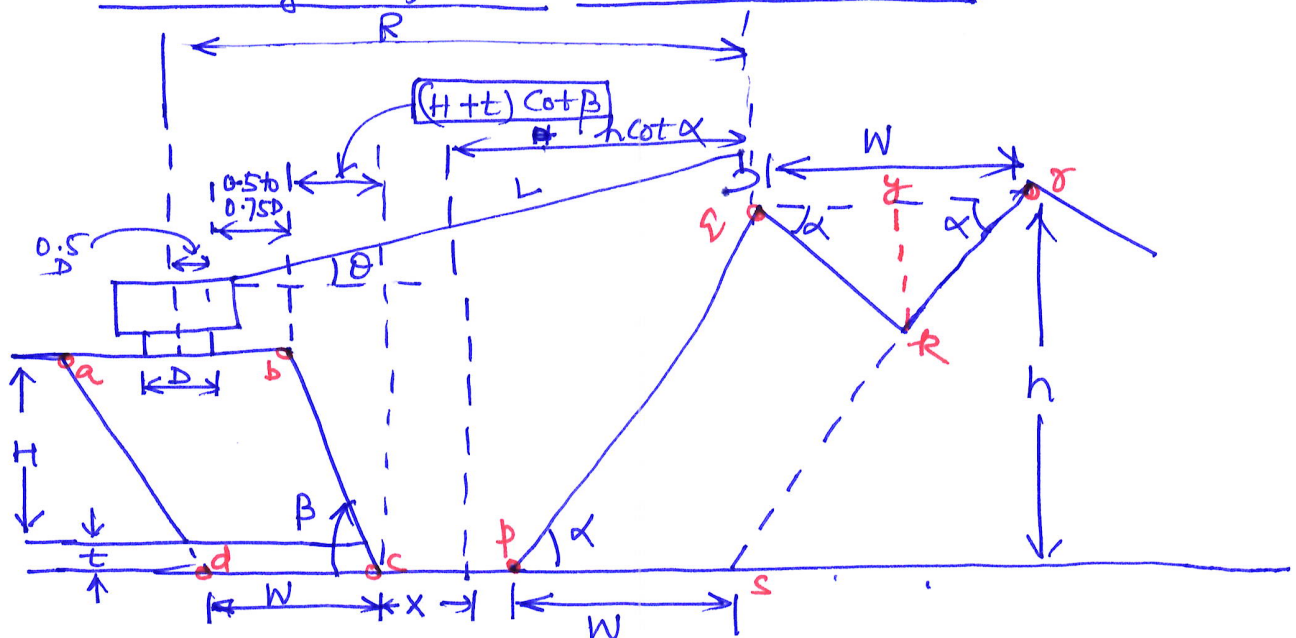
$$Q_{eff} = B_c \times B_f \times \frac{100}{100 + S} \times \frac{3600}{t_c \times F_s} \times E_s \quad (\text{m}^3/\text{hr})$$

L → (A2)

Equating Eqn ① & ②

$$B_c = \frac{(P \times S R) / (365 \times 24)}{\left(B_f \times \frac{100}{100 + S} \times \frac{3600}{t_c \times F_s} \right) \times E} \quad m^3 \quad \text{--- (A3)}$$

Balancing Diagram (Side Casting Diagram)



Here, Bench Parameters .

1. H = height of OB bench .
2. W = Width of cut .
3. t = thickness of coal seam .
4. β = highwall slope angle .

Dragline parameters

5. D = diameter of tub (chase) of dragline
6. L = Length of Boom
7. θ = Boom Angle .

Dump Parameters

8. $S = \% \text{ swell of material}$.
 $X = \text{clearance between high wall toe and spoil heap toe}$.
 $\alpha = \text{angle of slope of spoil heap}$.
 $h = \text{height of spoil heap}$.

The area (abcd) = insitu area of OB.

This area (abcd) is to be blasted and then it swells by a factor $\left(\frac{100+S}{100}\right)$. This swollen material is thrown in dump acquiring the shape (pqrs).

Now, $\text{area (abcd)} = H \times W$

$$\text{Area of dump (pqrs)} = H \times W \times \left(\frac{100+S}{100}\right)$$

$\longrightarrow \textcircled{1}$

$$\text{Now area (||}^m \text{pqrs)} = h \times W \longrightarrow \textcircled{2}$$

$$\begin{aligned} \angle \text{ Area } \Delta qRr &= 2 \times \text{Area } (\Delta qRy) \\ &= 2 \times \left[\frac{1}{2} \times \frac{W}{2} \times \frac{W}{2} \tan \alpha \right] \\ &= \frac{W^2}{4} \tan \alpha \longrightarrow \textcircled{3} \end{aligned}$$

But $\text{ar(pqrs)} = \text{ar}(\text{||}^m \text{pqrs}) - \text{ar}(\Delta qRr)$

Substituting Eqn (1) (2) & (3) in above relation.

Considering θ to be 35° to 39° (most common)

We get $L \approx R$

Here Eqn (5) is the reach of the Dragline.

MUF (Maximum Usefulness Factor) of Dragline.

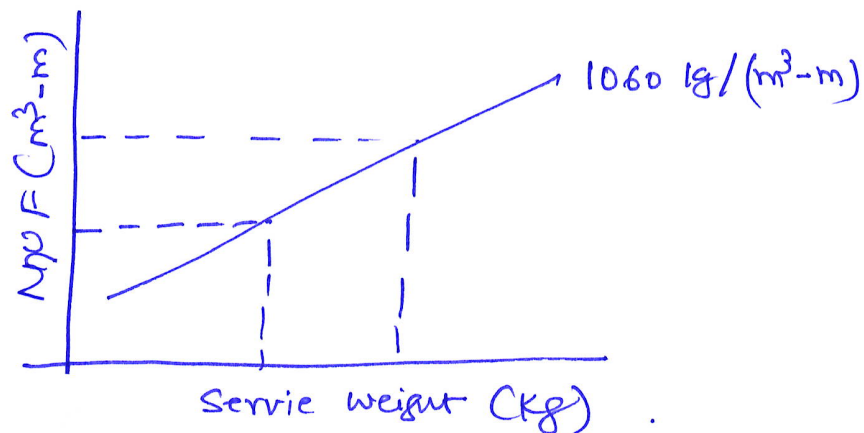
MUF is a measure of the stability of the dragline, and is given by

$$MUF = B_c \times R \approx B_c \times L$$

The MUF depends on the service weight of the machine.

Here Service weight = (Wt. of machine + Wt. of Counterbalancing Ballast)

If MUF Vs service weight of machine available throughout the world are plotted then the slope of the ~~line~~ regression line's inverse is given by $1060 \text{ kg}/(\text{m}^3\text{-m})$



Applicability of MUF.

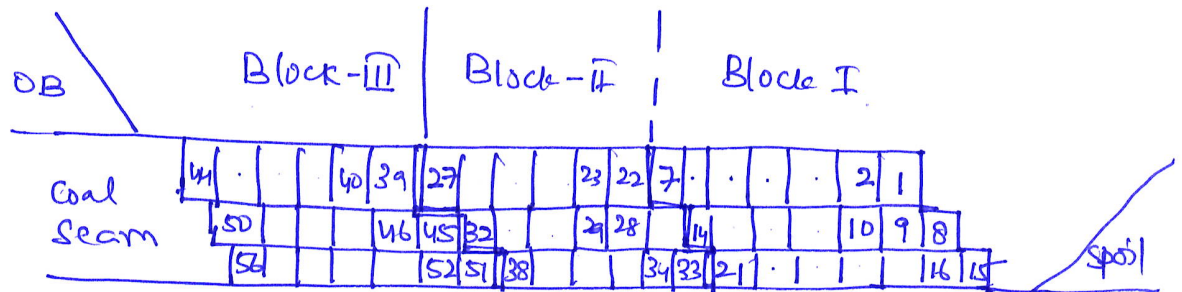
Once the required bucket capacity & reach for a particular mine is calculated based on the aforementioned eqs (A3) & eqn (5)

Then For the given MUF the ^{minimum} service weight is selected from the curve. ^

Thereafter few models from different manufacturers reference having service weight more than the minimum service weight are selected. From this the optimum one is chosen based on economic considerations.

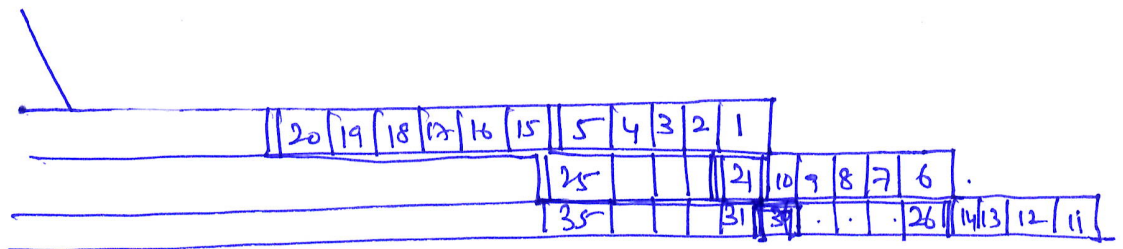
(ii) Block Mining Method.

In this method the bench width is divided into number of blocks and then the block from the edge side of the bench is extracted to the full height of the bench in number of slices following a sequence similar to that of wide bench method. After completion of one block, the next adjacent block is extracted till the whole bench is ~~started~~ excavated.



This method is adopted when the mineral bench is so wide that the extraction by wide bench method is not suitable.

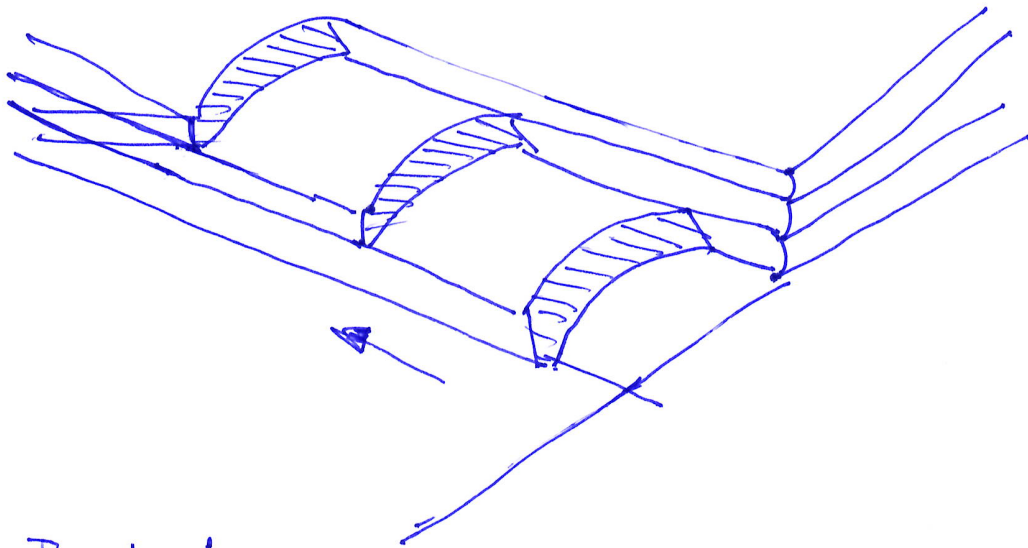
(iii) Stepped Cut Method:



This method is adopted when excess of moisture/water is present in the mineral bench that creates extraction problem. In this method also the bench is divided into suitable no. of blocks and the individual slices are so formed that ~~an~~ it forms a number of small steps on mineral bench. This facilitates drainage improving m/c efficiency.

- a). With the help of suitable sketches, describe the lateral block cut method of operation of bucket wheel excavator (BWE).
- b) Derive the theoretical hourly output for a bucket wheel excavator.
- c) Calculate the hourly productivity of the BWE from the following given data:
 - (i) No. of buckets on the wheel = 12
 - (ii) Individual bucket capacity = 1400 lit
 - (iii) Wheel Diameter = 10 m.
 - (iv) Cutting Speed at Knife edge = 2.2 m/s.
 - (v) % Swelling of material to be excavated } = 20 %.

The lateral block / Half Block Method of Working is shown in the figure below:



In the lateral block method, the machine travels parallel to the highwall with the crawlers offset. Long deep benches or terraces are cut along the highwall starting at the top. The angle of swing into the highwall is $10-15^\circ$ to the direction of travel. This generally requires a machine equipped with a long boom.

Productivity Calculation of BWE.

The calculation is based on how many buckets are discharged per unit time?

$$Q_{th} = I \times N_{BD} \times S_f \times 60 \frac{m^3}{hr} \quad \text{--- (1)}$$

I = individual bucket capacity
(given normally in litres it should be converted to m^3).

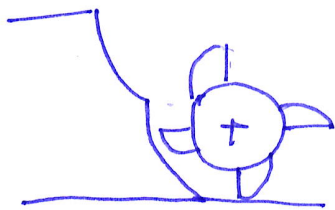
N_{BD} = No of Buckets discharged per unit time. (per minute)

$$S_f = \text{Swell factor} = \frac{100}{100 + S}$$

where S = % swell.

The N_{BD} is a function of V_c = Circumferential speed

B_N no of buckets on excavator & the Dia of the wheel of BWE.



Given:

B_N = No of buckets in BWE

V_c = Velocity of tip of bucket (m/s)

D = Dia of cutting wheel (m).

The problem is to find the no. of buckets discharged per unit time (minutes).

Now Time taken to discharge B_N Buckets

= time taken for full rotation.

$$= \frac{2\pi}{\omega} = \frac{2\pi}{V_c} \left(\frac{D}{2} \right) \quad \left[V_c = \left(\frac{D}{2} \right) \omega \right]$$

$$= \frac{\pi D}{V_c} \text{ seconds.}$$

∴ time taken to discharge B_N buckets

$$= \frac{\pi D}{V_c} \text{ sec}$$

$$= \frac{\pi D}{60 V_c} \text{ min.}$$

Now

$\frac{\pi D}{60 V_c}$ minutes are required to discharge B_N buckets.

$$\Rightarrow 1 \text{ minute is reqd to discharge} = \frac{B_N \times 60 \times V_c}{\pi D} \text{ buckets.}$$

∴ No of Buckets discharged/min

$$N_{BD} = \frac{60 V_c \times B_N}{\pi D} \quad \text{--- (2)}$$

∴ From (1) & (2) The theoretical hourly production of the BWE

$$\begin{aligned} Q_{th} &= I \times \frac{60 V_c \times B_N}{\pi D} \times S_f \times 60 \text{ (m}^3/\text{hr)} \\ &= \frac{I \cdot V_c \cdot B_N \cdot S_f}{\pi D} \times 3600 \text{ (m}^3/\text{hr)} \end{aligned}$$

(C) Given

$$B_N = 12.$$

$$\begin{aligned} I &= 1400 \text{ litres} \\ &= 1400 \times 10^{-3} \text{ m}^3 \\ &= 1.4 \text{ m}^3. \end{aligned}$$

$$D = 10 \text{ m}.$$

$$V_c = 2.2 \text{ m/s}$$

$$S_f = \frac{100}{100+20} = 0.833 = \left(\frac{5}{6}\right).$$

$$\begin{aligned} \therefore Q_{th} &= \frac{1.4 \times 2.2 \times 12 \times \left(\frac{5}{6}\right)}{\pi \times 10} \times 3600 \text{ m}^3 \\ &= 3529.42 \text{ m}^3/\text{hr}. \end{aligned}$$

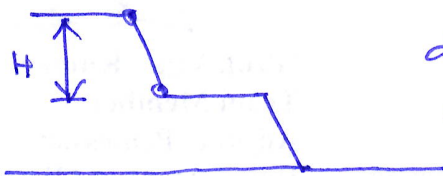
Ans

- (a) Enlist and discuss the geometrical parameters of production benches in Surface mine. How are these decided upon?
- (b) Enlist and discuss the types of failure that may occur in high wall slopes in Surface mines.

The followings are the geometrical parameters for production benches in Surface mines

- Bench height
- Bench width.
- Bench length.
- Bench face Slope angle.
- Bench floor slope angle.

a). Bench Height



The bench height is the vertical distance between the crest and toe of the bench.

DGMs Guideline for Bench Height

The bench height should not be more than

- Max digging ht of excavator excavating the bench in case of Hard formation.
- (Max digging ht of the excavator + 3m) in case of Soft formation.

Operational Guidelines for Bench Height

- optimum digging ht of excavator for efficient extraction.
- one pass drilling length should preferably be more than the bench height.
- blending requirements as dictated by Geological formations.

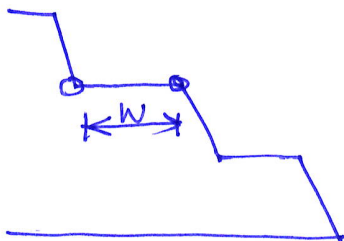
Equipment

1. Rope shovel & Dragline
2. Hydraulic excavator
3. Bucket wheel Excavator.

Guideline

- on the basis of optimum digging height of excavator.
- Max. digging height of the excavator.
- Depends on
- height of cutting boom pivot point
 - effective length of cutting boom including cutting wheel.
 - allowable max angle of inclination of cutting boom in vertical plane limited by the dynamic angle of repose of the cut material.

2 BENCH WIDTH:-



The bench width is the horizontal distance between the crest of the bench and the toe of the immediately upper bench.

DMS Guideline for Bench width.

The bench width shall be more than

- i) the bench height
- ii) Max width of largest m/c + 2m
- iii) $3 \times$ (width of largest dumper)
plying on the bench

Operational Guideline for Bench width.

The bench width (minimum) should be equal to

$$W = 0.8 \times H + \left(\begin{array}{c} \text{Dumper} \\ \text{width} \end{array} \right) + \left(\begin{array}{c} \text{Shovel} \\ \text{Dimension} \end{array} \right) + 4 \text{ (m)}$$

3. BENCH LENGTH:- It is the length (straight & curve) measured along the floor of the bench between two extremities.

Decided on Operational Requirements

The bench length is decided by frequency of primary blasting and the advancement of face along the length of bench per blast.

For Shovel dumper Combination, the min bench length may be calculated as follows

- Let
- (i) Frequency of blasting (primary) in the bench is once per week.
 - (ii) Volume of material to be handled per week $= V$
 - (iii) Height & width of benches respectively $= H \text{ \& } W$.

Then. Length 'l' of broken material required per week

$$l = \left(\frac{V}{H \times W} \right)$$

Then the Length 'L' of bench should preferably be

$$L = 3l = \frac{3V}{H \times W}$$

The length is kept 3 times because an additional length 'l' will be required for drilling while the blasted material is being handled by the excavator and another extra length 'l' for maintaining the equipment flexibility.

For Dragline Benches: Min width = 300 to 500m in view of marching requirements and Operational Convenience.

BWE Benches with Conveyor Combination decided upon.

- production requirements
- the method of work (full block / half block)
- frequency of Shifting & Shiftable Conveyors.
- Other Operational ~~and~~ Convenience.

Overall the max bench length is generally Selected on the basis of depositional feature and/or transport requirement.

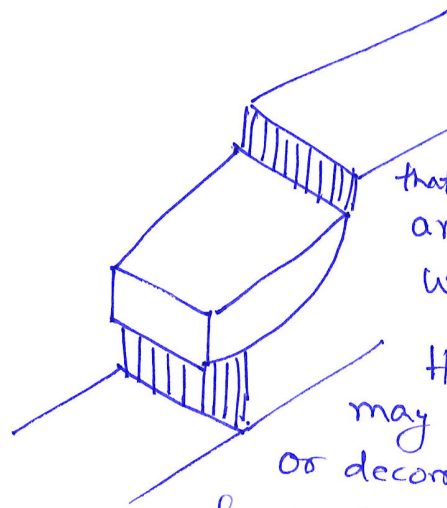
4. BENCH FACE SLOPE ~~ANGLE~~: It is a slope of a line (in vertical plane) that joins the toe and crest of the bench.

Selected on following criteria

- Geotechnical Stability of slope
- Inclined hole drilling requirement.
- Criteria of complete removal of blasted material by the excavator excavating the bench.

(5) Bench Floor SLOPE: Generally it is kept 1% to 2% of longitudinal slope to facilitate drainage of water to sump area.

(g) Single block / Circular Slip Path



Single block sliding failure that occurs along circular slip paths are commonly associated with soil slopes.

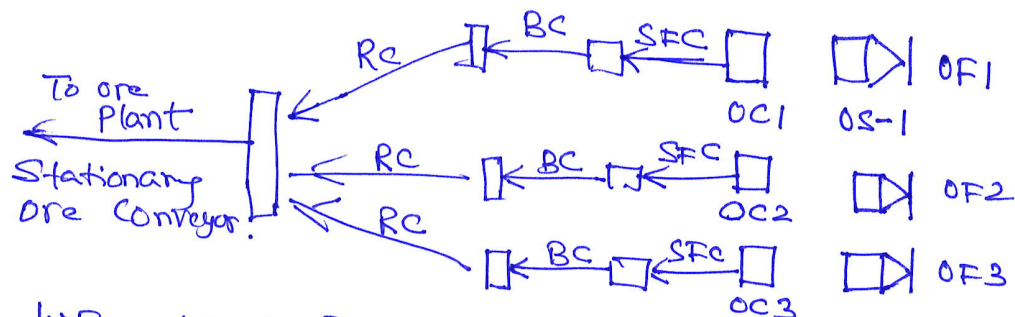
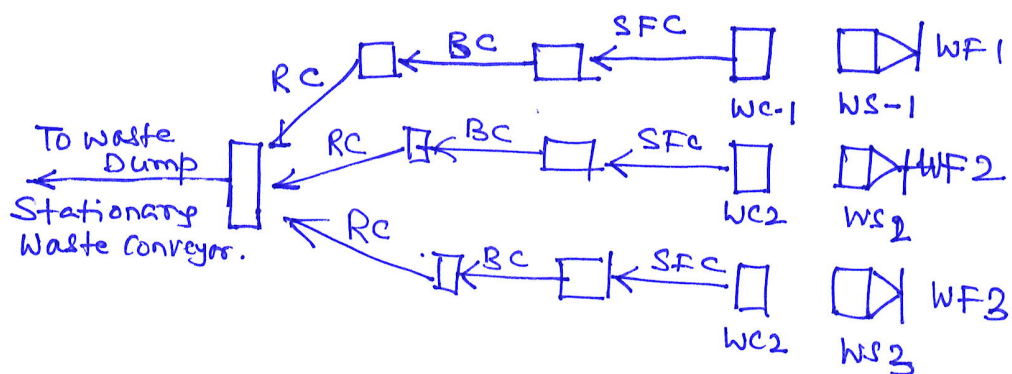
However, ~~S~~ circular slip failures may occur in highly weathered or decomposed rock masses / highly fractured rock such as clayey shales etc; or in poorly cemented sandstones.

7. Write short notes and (or) draw labeled diagram wherever necessary for the following

a) Schematic presentation of mobile in-pit crushing and conveying system.

b) Concept of tipping load in pay-loaders.

Ans a) Mobile in-pit Crushing System.



WF = Waste Face

OF = Ore face

WC = Waste Crusher

OC = Ore crusher

SFC = Shiftable Face Conveyor

BC = BENCH CONVEYOR

RC = Relocatable and/or Extendable Conveyor.

In mobile in-pit crushing system, the crusher is fully mobile and is located very close to the face.

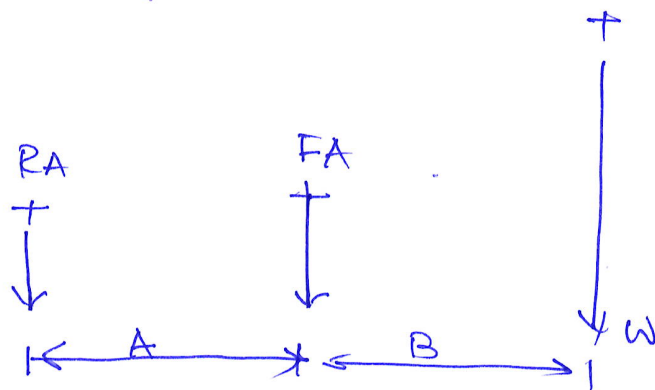
- The crusher is fed directly by an excavator.

- The crusher discharges onto the Shiftable face conveyor. Then to the Bench conveyor & to the RC & sent to destination by belt conveyors.

* The system is applicable for mine where only a few fast moving working faces exist and the excavator stay permanently at the same face.

CONCEPT OF TIPPING LOAD OF PAYLOADER

The minimum tipping load is the measure of the stability of FEL. The load distance representing FEL is shown below.



The tipping normally occurs at the fulcrum of Front Axle. Therefore the moment balance ~~diagram~~ equation is

$$A \times (RA) = W \times B$$

Where RA = machine load acting on the Rear axle.

W = dead weight of bucket + load on bucket

A = Distance between the axles of the machine

B = Distance between the FRONT AXLE & the Centre line of the bucket in fully extended position.

- Generally the minimum tipping load is about 70-80% of the machine weight depending on its configuration.
- Practically the max operating load $< 50\%$ of Min Tipping load.

The concept of minimum tipping load and maximum operating load is used to find out the suitable bucket size of a particular FEL for given job condition.